MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular Examination October 2025

Third Semester

MG3DSEFSQ202- Pre-Requisites in Food Industry

(2024 Admission Onwards)

Duration: 2 Hours **Maximum Marks: 50**

Part A
Multiple Choice Questions
Each question carries 1 mark each

1. **Use in potable water in food preparation by street vendors is a requirement of:**
   c). FSSAI Schedule 4 Part 1
2. **Proper stock rotation during food distribution ensures**
   a). FIFO and FEFO principles are followed
3. Select the international standard requires proper audit documentation and records in food industry
   d). ISO 22000/HACCP
4. **Select the material which is most suitable for meat contact surfaces**
   b). Stainless steel
5. **HACCP implementation without prerequisites programs become ineffective.**
   d). PRPs manage general hazards such as sanitation, pest control, layout, and hygiene
6. **FSSAI requires potable water in food premises to be:**
   d). Free from contaminants and meeting BIS standards
7. **In cold storage of food products, maintaining and appropriate temperature range ensures:**
   b). Increased product shelf life and safety
8. **A supervisor in the food industry should primarily focus on:**
   a). Training workers on hygiene and safety
9. **Select from options, which is NOT a sanitary requirement for milk storage**
   d). Storing milk directly on the floor in open vessels
10. **The placement of insect light traps in food premises should:**
    b). At entry points and near waste areas
11. **An ideal location for a food establishment is one that:**
    b). Avoids contamination sources like drains and waste areas
12. **Proximate analysis in food testing includes estimation of:**
    a). Protein, fat, carbohydrate, moisture and ash
13. **Personal hygiene is food handlers is important for:**
    a). Prevention contamination during food handling and processing

14. **Select the key requirement for personal hygiene of food handlers in catering establishment**
    b). Using clean protective clothing

15. **Storage area in food establishments must be:**
    a). Clean, dry, well- ventilated, and protected from pests

16. **Traceability in prerequisites programs help in:**
    a). Identifying the source of raw materials and finished products

17. **The proper method to prevent nail-related contamination during food preparation is:**
    b). Keeping nails short clean and unpolished

18. **The primary source of liquid food processing units include:**
    a). Raw materials using washing equipment

19. **Select from the option. Which type of flooring is required in a meat processing unit:**
    c). Smooth, washable and non- absorbent

20. **According to FSSAI Schedule 4 Part 3, waste from food processing units should be:**
    c). Collected, stored and disposed safely

Part B
Short Answer Questions
Answer any **6** questions
*Each question carries **5** marks*

21. Describe the personal hygiene practices that street food handlers must follow to prevent food contamination.

- Wash hands thoroughly before handling food and after restroom use.
- Wear clean clothes, apron, and head cover.
- Keep nails short and clean; avoid jewelry.
- Do not smoke, eat, or sneeze near food.
- Cover cuts or wounds and report illness to supervisor.

22. Explain the role of proper sealing techniques in maintaining the safety and quality of packaged food products.

- Prevents entry of contaminants, moisture, and air.
- Maintains internal pressure or vacuum to ensure product integrity.
- Extends shelf life and prevents spoilage.
- Provides tamper evidence and consumer safety.
- Maintains legal compliance and product traceability.

23. Describe the role of regular inspection and preventive maintenance schedules in ensuring hygienic premises in the food industry.

- Identifies equipment wear, damage, or leaks before breakdown.
- Prevents contamination due to malfunctioning machinery.
- Ensures compliance with hygiene and safety standards.
- Reduces downtime and improves productivity.
- Maintains records for verification and audits.

24. Write a short note on sanitation practices under GMP.

- Scheduled cleaning and disinfection of all areas.
- Use of approved detergents and sanitizers.
- Regular cleaning of equipment, utensils, and surfaces.
- Maintenance of cleaning logs and checklists.
- Verification of sanitation effectiveness through swab tests.

25. Explain the importance of monitoring water quality and sanitation in food processing environment.

- Ensures use of potable water meeting BIS/FSSAI standards.
- Prevents microbial and chemical contamination.
- Maintains hygiene in washing, processing, and cleaning.
- Regular testing and documentation support food safety.
- Poor water quality affects taste, colour, and shelf life of food.

26. Discuss the role FSSAI and license number in maintaining consumer confidence in India.

- FSSAI license mandatory for all food businesses.
- License number on label indicates legal compliance.
- Builds consumer trust and traceability.
- Helps in recall management and market surveillance.
- Ensures uniform national standards for safety and hygiene.

27. Describe the importance of proper procurement of raw materials as a prerequisite for unit operations in the food industry.

- Use of approved and reliable suppliers.
- Verification of raw material quality and specifications.
- Proper storage and transport to prevent spoilage.
- FIFO and FEFO principles followed.

- Reduces contamination risks and production losses.

28. Explain briefly solid waste management.

- Segregation of organic, recyclable, and hazardous waste.
- Use of color-coded bins for collection.
- Timely disposal through authorized agencies.
- Prevention of pest infestation and odour.

Part C

Essay

Answer any **2** questions

*Each question carries **10** marks*

29. Explain in detail about how can visitor become a source of contamination in food premises. Give examples.

- Visitors may introduce pathogens via clothing, footwear, or equipment.
- Contamination from respiratory droplets or poor hygiene.
- Unauthorized entry into processing areas can cause cross-contamination.
- Lack of awareness about GMP or hygiene rules.
- Preventive measures: restrict access, provide protective clothing, maintain visitor logs, ensure handwashing, and escort visitors.
- Example: Contamination of dairy product by visitor carrying pathogens.

30. Discuss the importance of designing food premises to allow easy cleaning and maintenance.

- Proper layout ensures smooth workflow and prevents cross-contamination.
- Smooth, washable walls and floors facilitate cleaning.
- Adequate lighting and ventilation prevent microbial growth.
- Slope in flooring for proper drainage.
- Avoids accumulation of dirt and pests.
- Reduces cleaning time, improves safety, and meets GMP standards.
- Easier inspection and maintenance enhance operational efficiency.
- Design supports segregation of raw and cooked zones.
- Example: Stainless steel equipment and tiled surfaces.
- Ensures compliance with FSSAI Schedule 4.

31. Discuss the significance of using smooth, washable surfaces in food equipment and containers.

- Prevents accumulation of dirt and microorganisms.
- Facilitates easy cleaning and sanitization.
- Increases equipment durability and prevents corrosion.
- Non-reactive materials prevent chemical contamination.
- Smooth surfaces reduce risk of biofilm formation.
- Helps maintain product quality and safety.
- Example: stainless steel, food-grade plastic.
- Meets FSSAI, BIS, and GMP hygiene requirements.
- Reduces cleaning time and resource use.
- Essential for compliance during food audits.

32. Discuss the role of management in implementing Good Manufacturing Practices (GMP) and food safety.

- Develops and enforces food safety policy and objectives.
- Provides training and awareness for employees.
- Allocates resources for sanitation and maintenance.
- Ensures documentation and internal audits.
- Monitors compliance with FSSAI and ISO 22000.
- Implements corrective and preventive actions (CAPA).
- Reviews performance through regular audits.
- Promotes food safety culture within organization.
- Liaises with regulatory authorities.